

# WIFI LAN AUDIT

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# LEARNING OBJECTIVES

Studying current infrastrucutre of the building

Collecting APs measurements

Ploting measurement in map and applying gradient

Studying and Suggesting improvement

# TOOLS:

1. **Autocad**: Sketching the map of building
2. **AdopePhotoshop**: Gradient
3. CMD : **netsh wlan** to get information and network configuration for APs
4. **WifInfoView** app measurements
5. AI to analyse our data and provide enhancement

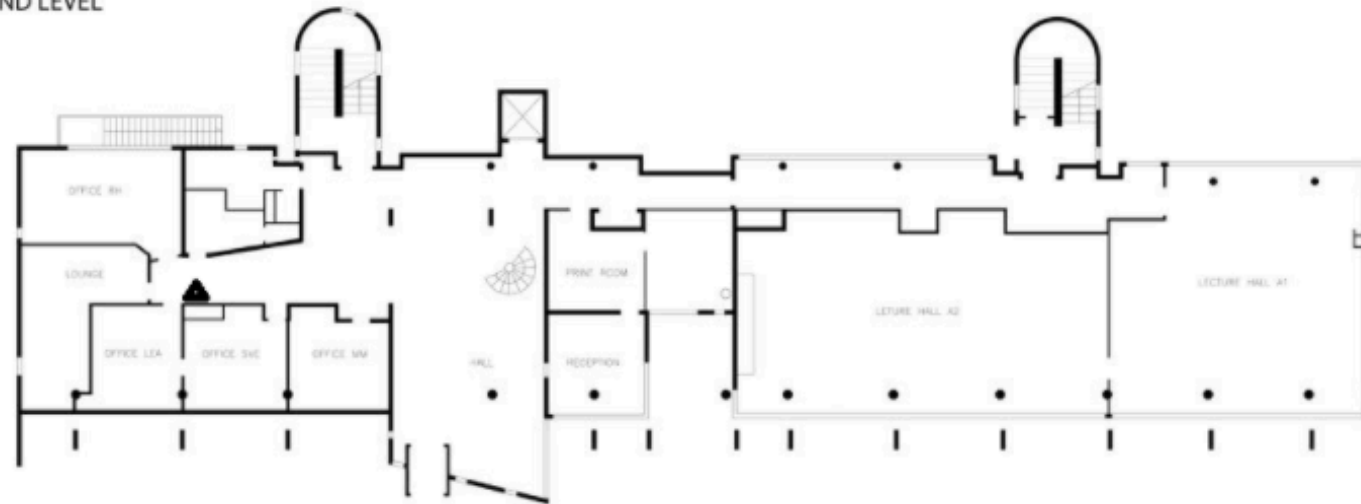


# CURRENT STATE

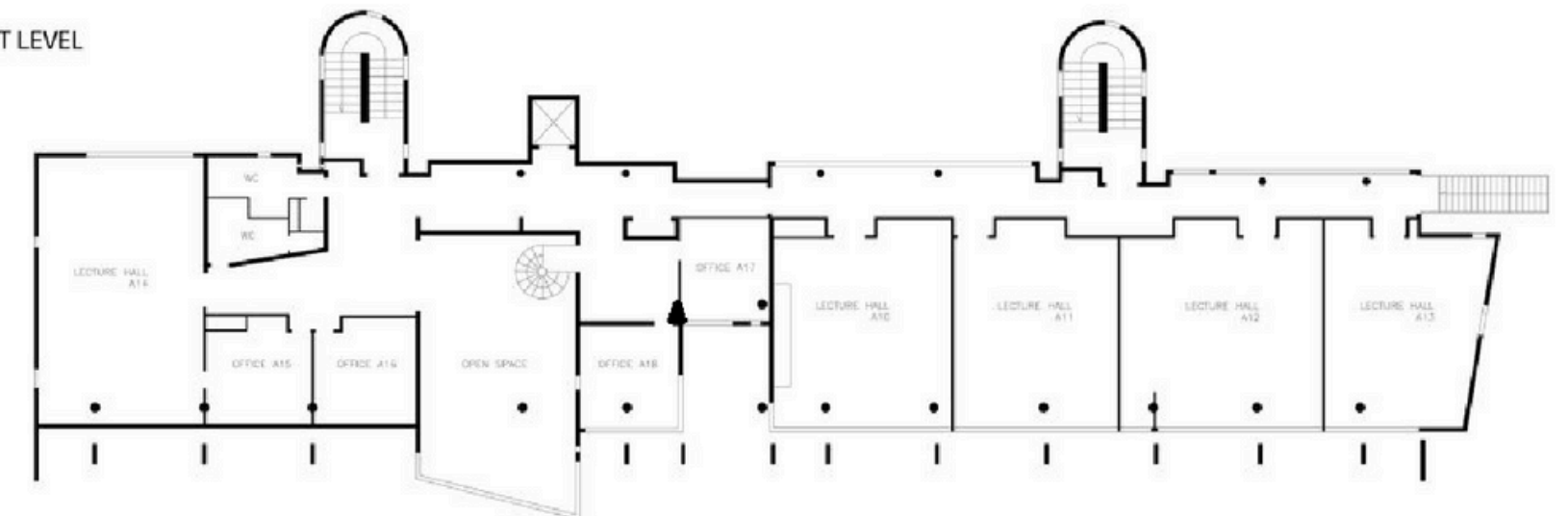
Building:STGI A

Number of floors:2 floors so we created 2 maps one for each floor

GROUND LEVEL



FIRST LEVEL



Number of Access Points:2 one in each floor



# APS CONFIGURATION

## AP1:Ground Floor

**SSID:**Eduroam

**MAC Address:**A4-BE-2B-C0-6C-90

**PHY Type:**802.11n/ac

**Frequency:**5.2Ghz-5.26Ghz

**Channel range:** 34-54

**Beacon packet:**202 bytes

**Security:**WPA2-EAP

**Channel width:**20Mhz

**Maximum speed:**867Mbps

**Theoretical subcarriers:**1333

**Practical Subcarriers:**1200

## AP2:First level

**SSID:**Eduroam

**MAC Address:**DC-99-14-80-C1-B0

**PHY Type:**802.11n/ac

**Frequency:**5.520Ghz-5.6Ghz

**Channel range:** 104-120

**Beacon packet:**205 bytes

**Security:**WPA2-EAP

**Channel width:**20Mhz

**Maximum speed:**867Mbps

**Theoretical subcarriers:**1333

**Practical Subcarriers:**1200

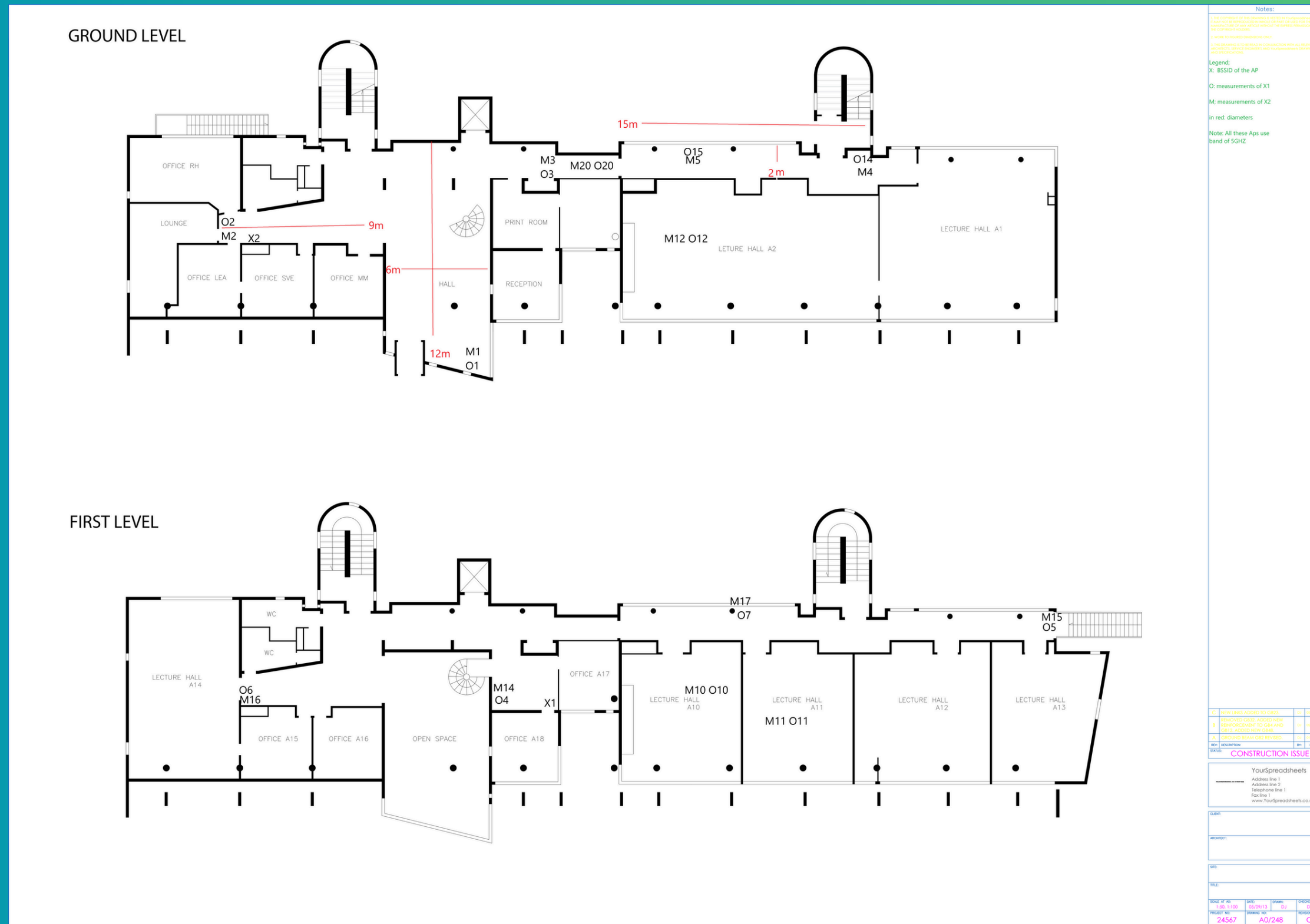


## Ground floor: From measurements

we found that AP1 which is also located as X2 in ground floor above secretary has stronger signal than AP2 which is in first floor so we used signal from AP1 to build heatmap for ground floor

**First floor:** Second AP X1 located in first floor has stronger signal than AP1 so for heatmap of first floor we used signal from AP2.

Measurements points noted with  
M(from AP1) and O (AP2)



# SAMPLE DATASET

| ReadingID                                | O1                | O2                | O3                | O4                | O5                | O6                | O7                | O10               | O11               | O12               | O14               | O15               |
|------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| SSID                                     | eduroam           | eduroam           | eduroam           | eduroam           | eduroam           | eduroam           | eduroam           | eduroam           | eduroam           | eduroam           | eduroam           | eduroam           |
| MAC Address                              | DC-99-14-80-C1-B0 | DC-99-14-80-C1-B0 | DC-99-14-80-C1-B0 | DC-99-14-80-C1-B0 | DC-99-14-80-C1-B0 | DC-99-14-80-C1-B0 | DC-99-14-80-C1-B0 | DC-99-14-80-C1-B0 | DC-99-14-80-C1-B0 | DC-99-14-80-C1-B0 | DC-99-14-80-C1-B0 | DC-99-14-80-C1-B0 |
| RSSI                                     | -65               | -47               | -67               | -67               | -45               | -76               | -53 ERROR         |                   | -73               | -85               | -87               | -88               |
| Signal quality(%)                        | 80                | 90                |                   | 75                | 95                | 50                | 88 ERROR          |                   | 60                | 26                | 22                | 20                |
| Frequency(GHZ)                           | 5.52              | 5.52              | 5.52              | 5.52              | 5.52              | 5.52              | 5.52 ERROR        |                   | 5.6               | 5.6               | 5.6               | 5.6               |
| Channel                                  | 104               | 104               | 104               | 104               | 104               | 104               | 104               | 104               | 104               | 104               | 104               | 104               |
| Theoretical data rate(Mbps)              | 90                | 133               |                   | 90                | 133               | 40                | 95 ERROR          |                   | 60                | 20                | 10                | 10                |
| Channel width(MHZ)                       | 20                | 20                | 20                | 20                | 20                | 20                | 20                | 20                | 20                | 20                | 20                | 20                |
| Bandwidth(GHZ)                           | 5                 | 5                 |                   | 5                 | 5                 | 5                 | 5                 | 5                 | 5                 | 5                 | 5                 | 5                 |
| Points on same coordinates(Ground floor) | O1M1              | O2M2              | O3M3              | O14M4             | O15M5             | O12M12            | O20M20            |                   |                   |                   |                   |                   |
| Average RSSI                             | -66.5             | -44.5             | -62               | -62               | -86               | -88               | -87               | -69.5             |                   |                   |                   |                   |
| Average data rate                        | 90                | 133               |                   | 95                | 15                | 15                | 12.5              | 65                |                   |                   |                   |                   |
| Points on same coordinates(First floor)  | O4M14             | O5M15             | O6M16             | O7M17             | O10M10            | O11M11            |                   |                   |                   |                   |                   |                   |
| Average RSSI                             | -57.5             | -72.5             |                   | -57               | -69               | -78.5             | -87.5             |                   |                   |                   |                   |                   |
| Average data rate                        | 86.5              | 65                |                   | 97.5              | 90                | 40                | 15                |                   |                   |                   |                   |                   |

Properties

SSID:

eduroam

MAC Address:

DC-99-14-80-C1-B0

PHY Type:

802.11n/ac

RSSI:

-65

Signal Quality:

80

Average Signal Quality:

81.0

Frequency:

5.520

Channel:

104

Information Size:

205

Elements Count:

17

Company:

HUAWEI TECHNOLOGIES CO.,LTD

Router Model:

Router Name:

Security:

WPA2-EAP

Cipher:

CCMP

Maximum Speed:

867 Mbps

Channel Width:

20 MHz

Channels Range:

102 - 106

BSS Type:

Infrastructure

WPS Support:

No

First Detection:

3/10/2025 3:18:18 PM

Last Detection:

3/10/2025 3:33:15 PM

Detection Count:

897

Start Time:

7/18/2024 1:36:36 PM

Minimum Signal Quality:

72

Maximum Signal Quality:

99

802.11 Standards:

802.11d/e/h/i

Connected:

No

Stations Count:

1

Channel Utilization:

0.8%

Country Code:

FR

Description:

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OK

Properties

SSID:

eduroam

MAC Address:

A4-BE-2B-C0-6C-90

PHY Type:

802.11n/ac

RSSI:

-68

Signal Quality:

72

Average Signal Quality:

74.7

Frequency:

5.240

Channel:

48

Information Size:

202

Elements Count:

16

Company:

HUAWEI TECHNOLOGIES CO.,LTD

Router Model:

Router Name:

Security:

WPA2-EAP

Cipher:

CCMP

Maximum Speed:

867 Mbps

Channel Width:

20 MHz

Channels Range:

46 - 50

BSS Type:

Infrastructure

WPS Support:

No

First Detection:

3/10/2025 3:18:18 PM

Last Detection:

3/10/2025 4:15:55 PM

Detection Count:

3439

Start Time:

7/18/2024 1:31:32 PM

Minimum Signal Quality:

18

Maximum Signal Quality:

99

802.11 Standards:

802.11d/e/i

Connected:

No

Stations Count:

1

Channel Utilization:

0.0%

Country Code:

FR

Description:

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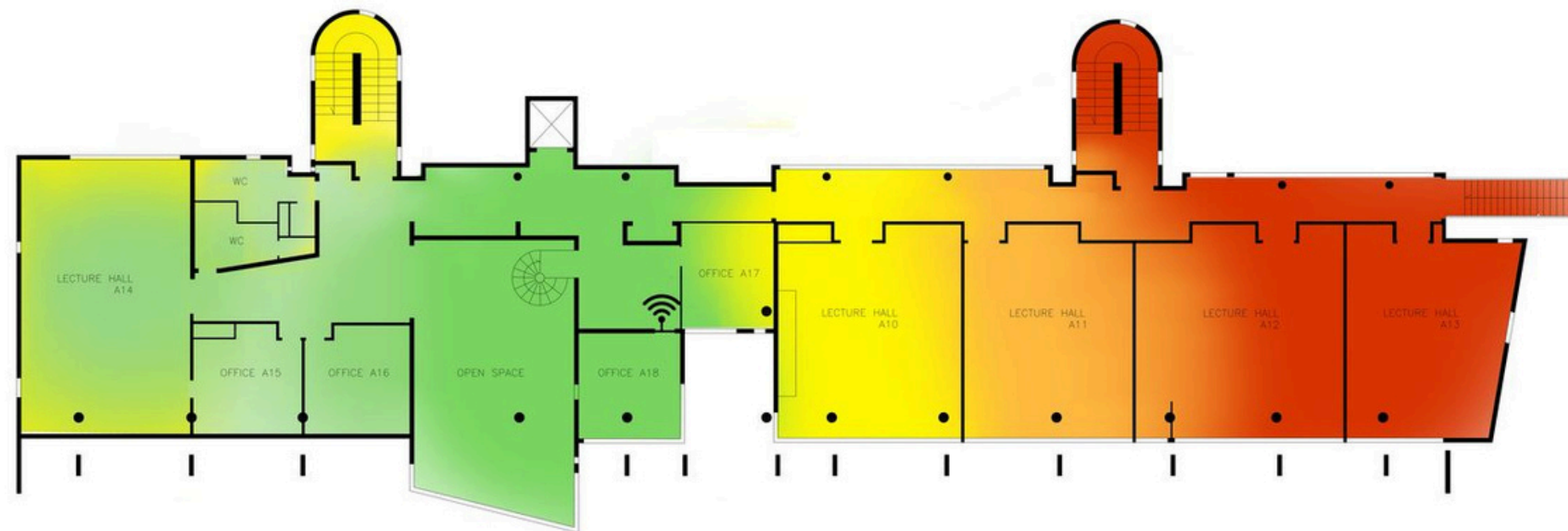
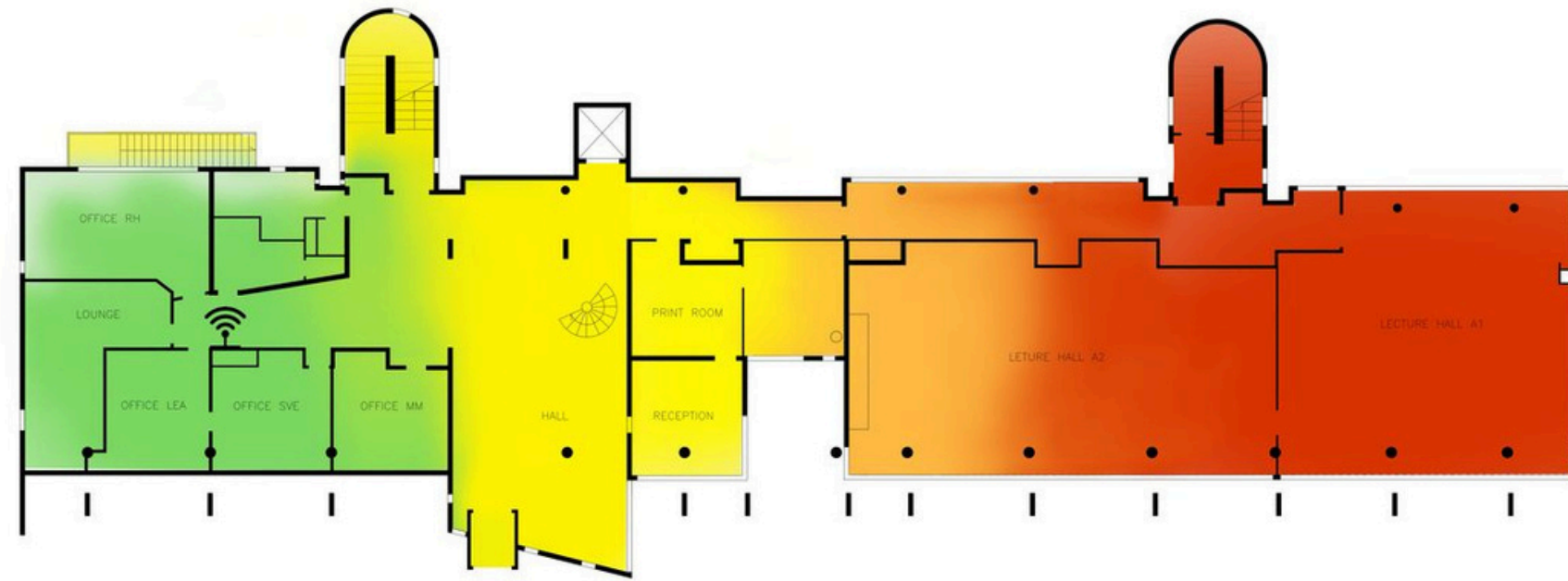
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OK

- This is our data that we have gathered from each measurement point from both APs into a table.
- Based on this RSS we calculated also data rate
- Based on strongest signal received comparing Mx and Ox points we generated Heatmap.



# HEAT MAP BASED ON SINGAL STRENGTH



## Legend

1. Wireless image show APs location
2. Green strong signal [-30,-50]
3. Yellow: Good overall [-50,-60]
4. Orange : Fair [-60,-70]
5. Red: Weak and very weak [-70,-80]
6. White spaces no signal

The heatmap is also valid for data rate as it depends on signal strength .

[illegible]



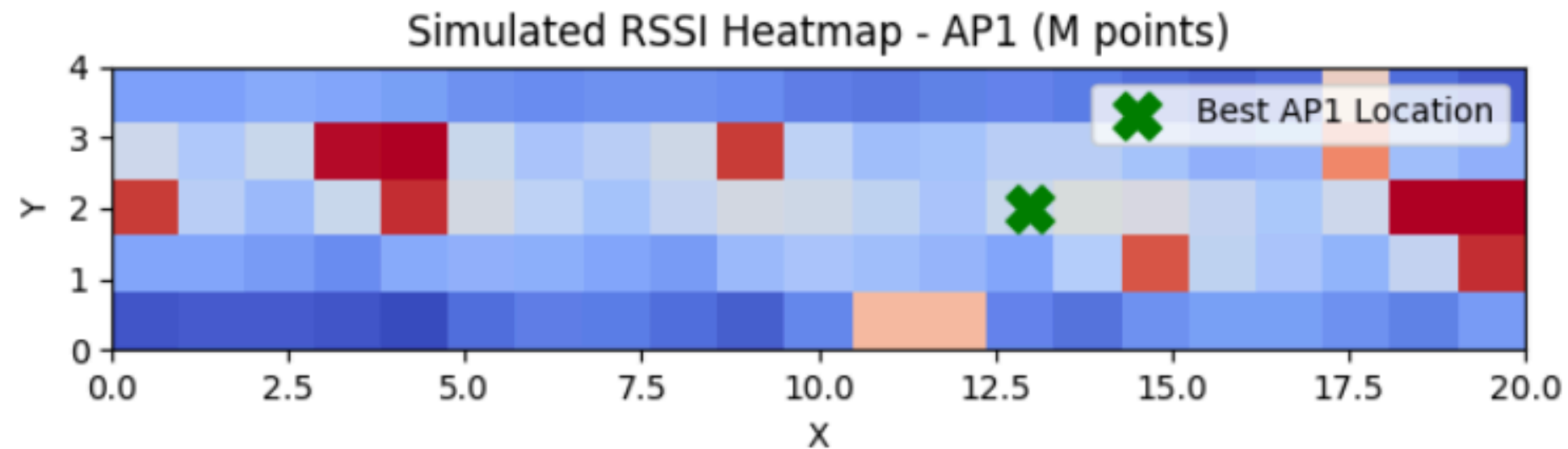
# SUGGESTIONS

- After analyzing the building's architecture and the wall materials, we found that the walls are very thick, significantly weakening the Wi-Fi signal and preventing it from reaching the opposite side of the building. To address this, we proposed installing two access points per floor—one on each side—instead of just one. This approach improves coverage and helps turn the previously red zones into green areas on the heatmap.
- Another suggestion is to move AP1 in central area like AP2 is to have LOF and to avoid thick walls.

# SUGGESTIONS

- In opposite part where is Red area of heatmap that corresponds to room classes we can use repeaters or mesh node that retransmit the signal.
- Upgrade to support WIFI 6 ,as currently they support only WIFI 4 and 5.Wifi 6 can optimize performance in dense environments.
- Check Antenna Orientation & Gain to use higher gain antenna and directional antennas to target low-signal areas.
- We can use AI to suggest AP placement,calculate RSS and to flags weak areas and recommends channel width or band changes.

# AI SUGGESTION

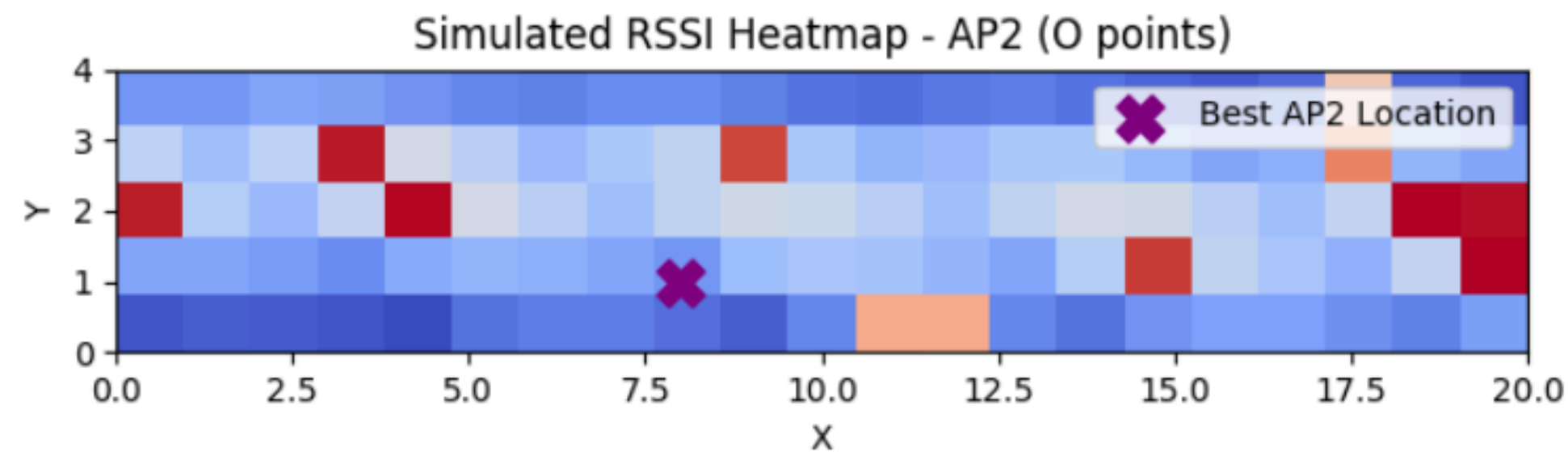


Best AP1 location: X=13, Y=2 (Avg RSSI  $\approx$  -30.74 dBm)

AP1 coverage ( $>$  -50 dBm): 68.57% of area

Best AP2 location: X=8, Y=1 (Avg RSSI  $\approx$  -30.51 dBm)

AP2 coverage ( $>$  -50 dBm): 68.57% of area

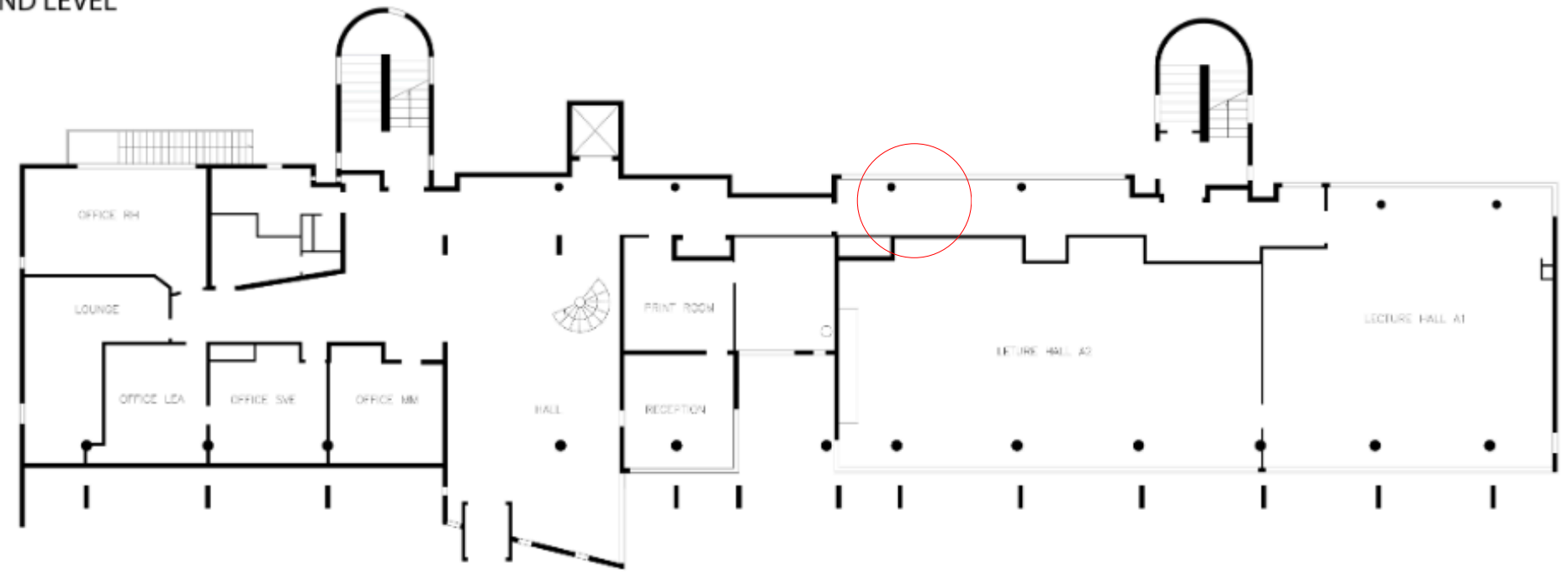


$$\text{RSSI} = \text{tx\_power} - 10 * n * \log_{10}(\text{distance} + \epsilon)$$

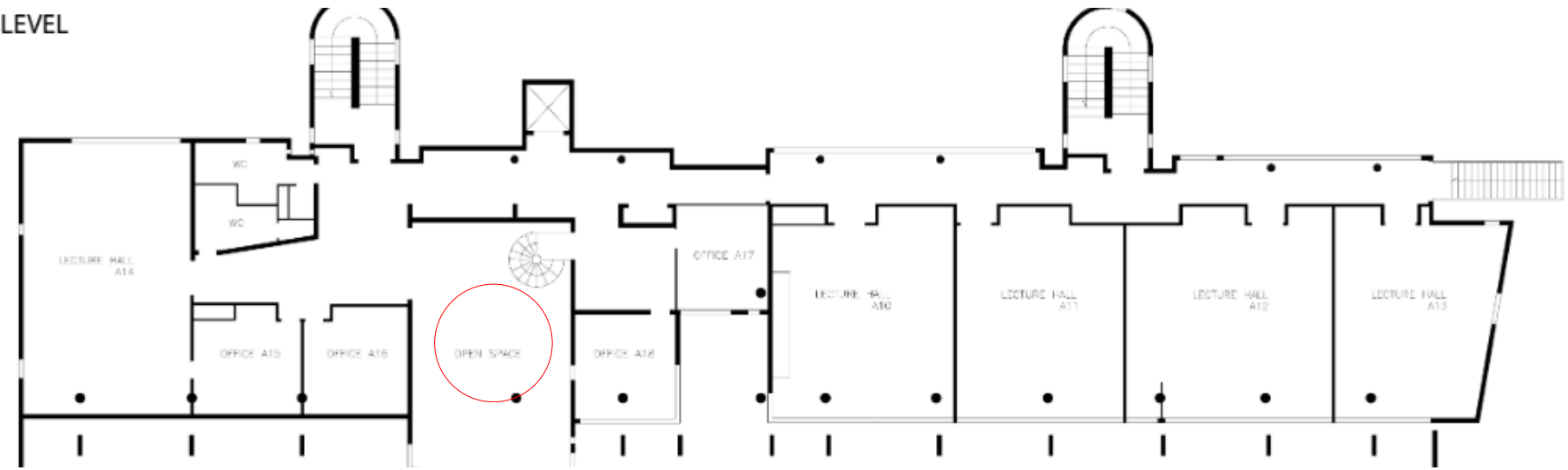
- tx\_power is the estimated RSSI at 1 meter
- n is the path loss exponent
- distance is how far each grid point is from the AP
- $\epsilon$  is a small constant to avoid division by zero

# AI RESULT

GROUND LEVEL



FIRST LEVEL



# Thank You!

<https://github.com/omarbkhalil/RadioNetworkWifiLanAudit>